PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI FACE ATTENDANCE SYSTEM

AN INTERNSHIP REPORT

Submitted by,

NIKHITHA LD – 20221CSE0696

Under the guidance of,

Mr.H.SENTHIL KUMAR

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

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BENGALURU

APRIL 2026

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"AI FACE ATTENDANCE SYSTEM"** is a bonafide work of **NIKHITHA.LD (20221CSE0696),** who has successfully carried out the internship work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-2026**.**

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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named NIKHITHA LD, hereby declare that the internship work titled **"AI FACE ATTENDANCE SYSTEM"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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PLACE:BENGALURU

DATE: 22-April-2026

INTERNSHIP COMPLETION CERTIFICATE

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रेल मंत्रालय / Ministry of Railways

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CERTIFICATE

This is to certify that **Ms. NIKHITHA L D, (20221CSE0696)**, B.Tech (Computer Science & Engineering) Student of Presidency University, has underwent Internship on **"Full Stack Web Development - AI Face attendance System" at Rail Wheel Factory, Yelahanka, Bangalore** for a period of 03 months i.e., 06th January 2026 to 06th April 2026.

The student has shown keen interest in learning during the period she was with us for the Internship. We found her Enthusiastic, Diligent, Hard Working and Creative.

We wish a bright future and success in all her endeavors.

(K.S.KIRAN)

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NIKHITHA LD

ABSTRACT

The internship project titled **"AI Face Attendance System"** aims to automate the traditional attendance system using Artificial Intelligence and Computer Vision technologies. Manual attendance systems are time-consuming, prone to errors, and allow proxy attendance. This project provides an efficient and automated solution using face recognition.

The system is developed using **React.js (Frontend)**, **Flask (Backend)**, and **Insight Face with OpenCV (AI Model)**. It captures real-time video from a webcam, detects faces, and recognizes individuals by comparing them with stored datasets. Once identified, attendance is automatically recorded in the database with date and time.

The system ensures accuracy, efficiency, and security. It reduces manual effort, prevents proxy attendance, and provides real-time attendance tracking. The project also includes a web-based dashboard for managing users and viewing attendance reports.

This internship provided hands-on experience in **full-stack development, AI integration, and real-time system implementation**, enhancing both technical and problem-solving skills.

The face recognition process is powered by advanced deep learning models from **InsightFace**, which provide high accuracy in detecting and recognizing faces even under varying lighting conditions and angles. The system uses facial embeddings to uniquely identify individuals, making it more reliable compared to traditional methods. The integration of **OpenCV** enables real-time video capture and face detection, ensuring smooth and continuous operation.

The attendance data is securely stored in a **SQLite database**, which maintains records such as user identity, date, and time of attendance. The system also provides functionalities to view attendance history, generate reports, and manage user data through an intuitive dashboard.

From a performance perspective, the system is optimized to handle multiple users and real-time processing without significant delay. Various testing methods were applied to ensure the reliability and accuracy of the system, including functional testing, integration testing, and real-time camera testing.

Furthermore, the system is cost-effective as it utilizes open-source technologies and does not require expensive hardware like biometric scanners. It is highly adaptable and can be deployed in various environments such as educational institutions, offices, and industries.

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ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface

Chapter 1

INTRODUCTION

The **AI Face Recognition Attendance Management System** is an advanced solution designed to automate and improve the traditional attendance process using Artificial Intelligence (AI) and Computer Vision technologies. In many organizations such as colleges, schools, and industries, attendance is still recorded manually or using basic biometric systems. These traditional methods are often time-consuming, prone to human errors, and vulnerable to proxy attendance.

With the rapid advancement of technology, there is a growing need for intelligent systems that can automate routine tasks efficiently. Face recognition is one of the most powerful applications of AI, which enables the identification of individuals based on their unique facial features. This project leverages face recognition technology to build a smart attendance system that automatically marks attendance without human intervention.

The system captures live video through a webcam, detects human faces using OpenCV, and recognizes them using deep learning models provided by InsightFace. Once a face is recognized, the system records attendance in the database with accurate date and time. This eliminates the need for manual entry and significantly reduces the chances of errors or manipulation.

The application is developed using modern web technologies. The frontend is built using React.js, which provides an interactive and user-friendly interface. The backend is implemented using Flask, which handles API requests, data processing, and communication between modules. The system follows a client-server architecture, ensuring scalability and efficient data flow.

This project was developed during the internship at **Rail Wheel Factory (RWF), Bengaluru**, where the focus was on applying AI and full-stack development skills to solve real-world problems. The system is designed to be efficient, accurate, and easy to use, making it suitable for deployment in various environments.

The system works by capturing an image or video of a person's face through a camera, then comparing it with stored facial data in a database. Using advanced techniques from Artificial Intelligence and Computer Vision, the system can detect, recognize, and verify a person's identity in real time.

This technology is widely used in schools, colleges, offices, and organizations to ensure accurate and efficient attendance tracking. It reduces the chances of proxy attendance, improves security, and saves time by automating the entire process.

AI-based attendance systems often rely on powerful algorithms such as Machine Learning and Deep Learning to continuously improve recognition accuracy, even under different lighting conditions, angles, and facial expressions.

An **AI Face Attendance System** is an intelligent, automated solution that leverages Artificial Intelligence to manage attendance with high accuracy and efficiency. It replaces traditional attendance methods by using a person's face as a unique identity, making the process faster, contactless, and more secure.

The AI Face Attendance System is a smart and efficient solution that modernizes attendance tracking. By combining Artificial Intelligence with Computer Vision, it provides a fast, reliable, and secure alternative to traditional methods. As technology evolves, these systems will become even more accurate, affordable, and widely adopted.

1.1 Background

Attendance management is an essential activity in educational institutions and workplaces. Traditionally, attendance is recorded manually using registers or through basic systems such as ID cards or biometric devices. However, these methods have several drawbacks:

- Manual attendance is time-consuming and inefficient
- High chances of proxy or fake attendance
- Difficult to maintain and manage records
- Requires additional manpower
- Limited accuracy and reliability

With the emergence of Artificial Intelligence, face recognition has become a reliable and efficient method for identification. AI-based systems can automatically detect and recognize individuals, making them suitable for attendance management.

This project utilizes AI and Computer Vision to overcome the limitations of traditional attendance systems and provide a modern, automated solution.

With the help of computer vision and machine learning, this system captures and analyzes facial features in real time, identifies individuals, and records attendance automatically. It ensures higher accuracy, improved security, and reduced human effort.

The rapid development of artificial intelligence has significantly transformed traditional systems into smart and automated solutions. One such advancement is the use of facial recognition technology for attendance management. Conventional attendance methods, such as paper registers and biometric fingerprint systems, are often inefficient, time-consuming, and prone to errors like proxy attendance or device misuse.

Facial recognition technology works by capturing an image of a person's face and comparing it with stored data using machine learning algorithms. Each face has unique features such as the distance between eyes, nose shape, and jawline structure, which are analyzed to identify individuals accurately. This makes the system more reliable and secure compared to traditional methods.

This system eliminates the need for physical contact, reduces manual effort, and ensures accurate and secure attendance tracking. It is widely applicable in schools, colleges, offices, and other organizations where efficient attendance management is required.

1.2 Statistics of Internship

The internship provided practical exposure to real-world application development and AI integration.

Key details:

- **Internship Organization:** Rail Wheel Factory (RWF), Bengaluru
- **Duration:**Approximately 12 - weeks
- **Domain:** Artificial Intelligence & Full Stack Development
- **Technologies Used:** React.js, Flask, OpenCV, InsightFace, SQLite
- **Modules Developed:**
    1. Face Registration
    2. Dataset Collection
    3. Model Training
    4. Live Face Recognition
    5. Attendance Logging

The internship helped in gaining hands-on experience in designing, developing, and deploying a real-time system.

1.3 Prior Existing Technologies

Before the development of AI-based attendance systems, several traditional methods were used:

1. Manual Attendance System

- Uses registers or paper-based records
- Time-consuming and error-prone
- Difficult to maintain records

2. RFID-Based Systems

- Uses ID cards for attendance
- Can be misused by sharing cards

3. Biometric Systems

- Uses fingerprint or iris recognition
- Requires expensive hardware
- Hygiene concerns in shared usage

Limitations of Existing Systems:

- Lack of automation
- Possibility of fraud (proxy attendance)
- High maintenance cost
- Limited scalability

- Time-consuming and inefficient
- High chances of **human error**
- Easy to manipulate (proxy attendance)
- Difficult to maintain large records
- Students/employees can **swap cards** (proxy issue)
- Cards can be **lost, damaged, or stolen**
- Additional cost for issuing and maintaining cards
- Requires physical interaction (less hygienic)

These limitations highlight the need for an intelligent and automated system.

1.4 Proposed Approach

The proposed system is an **AI-based Face Recognition Attendance System** that automates the attendance process.

Working Approach:

1. Capture live video using webcam
2. Detect faces using OpenCV
3. Extract facial features using InsightFace
4. Compare with stored dataset
5. Identify the person
6. Mark attendance automatically

Advantages:

- Fully automated system
- High accuracy
- Real-time processing
- No physical contact required
- Secure and reliable

1.5 Objectives

The main objectives of the project are:

- To develop an AI-based attendance system
- To automate the attendance process
- To reduce manual effort and human errors
- To prevent proxy attendance
- To provide real-time attendance tracking
- To create a user-friendly web interface
- To improve efficiency and productivity

1.6 SDGs (Sustainable Development Goals)

The project aligns with the following United Nations Sustainable Development Goals (SDGs):

Fig : 1.6 SDGs (Sustainable Development Goals)

1.7 Overview of Internship Report

This report is structured into multiple chapters:

- **Chapter 1:** Introduction
- **Chapter 2:** Literature Survey
- **Chapter 3:** Methodology
- **Chapter 4:** Project Management
- **Chapter 5:** Analysis and Design
- **Chapter 6:** Implementation
- **Chapter 7:** Evaluation and Results
- **Chapter 8:** Social and Ethical Aspects
- **Conclusion & References**

This structured report provides a comprehensive understanding of the project and the work carried out during the internship.

Chapter 2

LITERATURE SURVEY

The literature survey provides an overview of existing technologies, research work, and systems related to attendance management and face recognition. It helps in understanding the evolution of attendance systems, their limitations, and the need for advanced AI-based solutions. This chapter focuses on traditional attendance systems, biometric technologies, and modern AI-based face recognition systems.

2.1 Introduction to Attendance Systems

Attendance management systems are used to record and track the presence of individuals in organizations such as schools, colleges, and workplaces. Over time, these systems have evolved from manual methods to automated digital solutions.

Traditional systems were mainly paper-based, where attendance was recorded manually. With technological advancements, automated systems such as RFID and biometric systems were introduced. However, these systems still have limitations in terms of efficiency, cost, and security.

Attendance systems are used to record and manage the presence of individuals in organizations such as schools, colleges, and workplaces. Traditionally, attendance was tracked manually using registers or sheets, where a teacher or manager would mark each person as present or absent. While simple, this method is time-consuming, prone to human error, and vulnerable to manipulation (like proxy attendance).

To improve accuracy and efficiency, modern attendance systems have evolved into automated solutions. These include technologies like RFID cards, biometric systems (fingerprint or iris scanning), and digital logging systems. Among these, biometric systems became popular because they uniquely identify individuals based on physical traits, reducing chances of fraud.

Recent developments in Artificial Intelligence and Computer Vision have enabled the creation of smart attendance systems that can automatically recognize individuals using facial features.

2.2 Traditional Attendance Systems

2.2.1 Manual Attendance System

The manual attendance system involves recording attendance using registers or sheets.

Features :

- Uses camera + AI algorithms
- Identifies faces in real-time
- Stores data digitally

- No physical contact required

Advantages:

- Simple to implement
- No technical knowledge required
- Fast and automatic process
- High accuracy (if properly trained)
- Prevents proxy attendance
- Saves time and effort
- Easy data management (reports, logs)

Disadvantages:

- Time-consuming
- High chances of human error
- Possibility of proxy attendance
- Difficult to maintain records
- Initial setup cost is high
- Requires electricity and system maintenance
- Privacy concerns
- Can fail with poor lighting or face changes

2.2.2 RFID-Based Systems

RFID (Radio Frequency Identification) systems use ID cards to mark attendance.An **AI Face Attendance System** and an **RFID-based attendance system** are two modern methods used to automate attendance tracking. The AI face system uses cameras and facial recognition technology to identify individuals and mark attendance automatically without physical contact, making it highly secure and preventing proxy attendance. In contrast, an RFID system uses a card or tag that users must scan on a reader, which makes it simple, fast, and cost-effective but allows the possibility of proxy attendance if cards are shared. While AI-based systems offer higher accuracy and advanced security features, they require more complex setup and higher cost, whereas RFID systems are easier to implement and maintain. Both systems improve efficiency compared to manual attendance, and in some cases, a combination of both can provide better reliability and flexibility.

Advantages:

- Faster than manual systems
- Easy to use
- Fast and Contactless Attendanc
- Cost-Effective
- Reliable and Accurate
- Works in All Lighting Conditions
- Quick Data Processing
- Low Power Consumption
- Scalable System

Disadvantages:

- Cards can be shared (proxy attendance)
- Requires additional hardware
- Risk of card loss or damage
- Requires Physical Interaction
- Limited Security
- Maintenance Cost

2.2.3 Biometric Systems

Biometric systems use physical characteristics such as fingerprints or iris scans. An **AI Face Attendance System** is a modern biometric solution that uses facial recognition technology to automatically record and manage attendance. Unlike traditional methods such as manual registers or fingerprint scanners, this system captures an individual's face through a camera and compares it with stored data using artificial intelligence algorithms. It ensures high accuracy, reduces proxy attendance, and improves security by identifying unique facial features. The system works in real time, making it efficient for schools, colleges, and workplaces. Additionally, it is contactless, hygienic, and minimizes human effort, making it a reliable and advanced alternative to conventional biometric systems.

As part of broader **biometric systems**, AI face attendance belongs to the field of Biometrics, which focuses on identifying individuals based on physical or behavioral characteristics. Other common biometric methods include fingerprint recognition, iris scanning, voice recognition, and Facial Recognition Technology. Among these, facial recognition is gaining popularity due to its non-intrusive and user-friendly nature. AI algorithms, especially deep learning models like convolutional neural networks (CNNs), enhance accuracy by learning facial patterns even under varying lighting conditions, angles, and expressions.

Advantages:

- More secure than RFID
- Unique identification
- Accuracy & Security
- Fast & Contactless Operation
- Automation & Efficiency
- Remote Monitoring & Integration
- Cost-Effective in the Long Run

Disadvantages:

- Expensive hardware
- Hygiene issues (touch-based systems)
- Slow when handling large groups
- Contactless & Hygienic
- High Accuracy
- Prevents Proxy Attendance
- Fast and Time-Saving
- Automated Record Keeping
- Real-Time Monitoring

2.3 Face Recognition Technology

Face recognition is a biometric technology that identifies individuals based on their facial features. It is widely used in security systems, surveillance, and attendance management.

The core of face recognition works through deep learning models such as Convolutional Neural Networks, which are trained on large datasets to accurately distinguish between different faces even under varying lighting conditions, angles, and expressions. Once a match is found, the system automatically marks attendance with a timestamp, eliminating the need for manual registers or biometric fingerprint devices. This improves efficiency, reduces proxy attendance, and enhances security.

Additionally, modern AI-based attendance systems often integrate cloud storage, real-time monitoring, and reporting dashboards, making it easy for administrators to track attendance patterns and generate reports. However, these systems also raise concerns about privacy, data security, and ethical use, as facial data is sensitive and must be protected using encryption and secure storage methods. Overall, AI face attendance systems represent a smart, contactless, and efficient approach to attendance management in the digital era.

The captured image is processed using algorithms that detect faces, extract unique facial features (like the distance between eyes, nose shape, and jawline), and convert them into a digital format known as a facial embedding. This embedding is then compared with a stored database of registered individuals to verify identity.

The process of face recognition includes:

1. Face Detection
2. Feature Extraction
3. Face Matching

Face recognition systems use machine learning and deep learning algorithms to achieve high accuracy.

2.4 AI-Based Face Recognition Systems

With the advancement of Artificial Intelligence, face recognition systems have become more accurate and efficient.

An AI Face Attendance System is an advanced application of Artificial Intelligence and Computer Vision that automates attendance marking by recognizing human faces instead of relying on manual methods like registers, ID cards, or biometric fingerprint scanners. The system works by capturing real-time images or video through a camera, detecting faces within the frame, and then comparing those faces with a pre-stored database using powerful algorithms such as Deep Learning and Neural Networks. Once a match is found, the system automatically records the attendance along with the time and date, making the process fast, accurate, and contactless.

This system typically involves multiple stages, including face detection, feature extraction, and face recognition. In the detection phase, the system identifies human faces in an image using techniques like Haar cascades or deep learning-based models. In the feature extraction stage, unique facial features such as the distance between eyes, nose shape, and jawline are

converted into numerical data called embeddings. Finally, during recognition, these embeddings are compared with stored data to identify the individual. Modern systems often use advanced frameworks like OpenCV and libraries such as TensorFlow or PyTorch to achieve high accuracy and efficiency.

AI-based face recognition attendance systems offer several advantages, including eliminating proxy attendance, reducing human errors, and improving security. They are widely used in schools, colleges, offices, and organizations to maintain reliable attendance records. Additionally, these systems can be integrated with cloud storage and mobile applications for real-time monitoring and reporting. However, they also raise concerns related to privacy, data security, and ethical use, as facial data is highly sensitive. Proper encryption, secure storage, and compliance with data protection regulations are essential to ensure safe implementation of these systems.

Key Features:

- Automatic face detection
- Real-time recognition
- High accuracy using deep learning
- No physical contact required
- Detects human faces from camera input in real-time
- Matches detected faces with stored database
- Uses advanced AI models like deep learning for high accuracy
- No need for fingerprint or ID cards
- Hygienic and safe (especially useful post-pandemic)
- Reduces proxy attendance (no “buddy punching”)

AI-based systems use neural networks to extract unique facial features and compare them with stored data.

2.5 Computer Vision in Attendance Systems

Computer Vision is a field of AI that enables machines to interpret visual data.

An AI Face Attendance System is an advanced application of Computer Vision that automates the process of marking attendance by recognizing and verifying human faces. Unlike traditional attendance methods such as manual registers, RFID cards, or biometric fingerprint systems, this system uses cameras and intelligent algorithms to detect, capture, and identify faces in real time. It typically works by first enrolling individuals into a database, where their facial features are extracted and stored as numerical representations (feature vectors). When a person appears in front of the camera, the system detects the face, compares it with stored data, and marks attendance if a match is found.

The core technologies behind AI face attendance systems include Face Detection and Face Recognition. Face detection identifies the presence and location of a face within an image or video frame, while face recognition analyzes unique facial features such as the distance between eyes, nose shape, and jawline to determine identity. Modern systems often rely on deep learning models, especially Convolutional Neural Networks (CNNs), which improve accuracy even under varying lighting conditions, angles, and facial expressions.

In attendance systems, computer vision is used for:

- Capturing images/video
- Detecting faces
- Tracking movement
- Recognizing individuals

Libraries like **OpenCV** play a major role in implementing computer vision functionalities.

2.6 Deep Learning in Face Recognition

Deep learning techniques improve the accuracy of face recognition systems.

An AI Face Attendance System uses deep learning–based face recognition to automatically identify and record the presence of individuals without manual intervention. It typically relies on models such as Convolutional Neural Networks (CNNs) to extract unique facial features from images or live video streams. These features are converted into numerical embeddings and compared with stored data to accurately recognize each person. Deep learning improves the system's performance by handling variations in lighting, facial expressions, and angles, making it more reliable than traditional methods. Such systems enhance security, reduce proxy attendance, and save time in environments like schools, offices, and organizations by providing a fast, contactless, and efficient attendance tracking solution.

Common Techniques:

1. Convolutional Neural Networks (CNN)

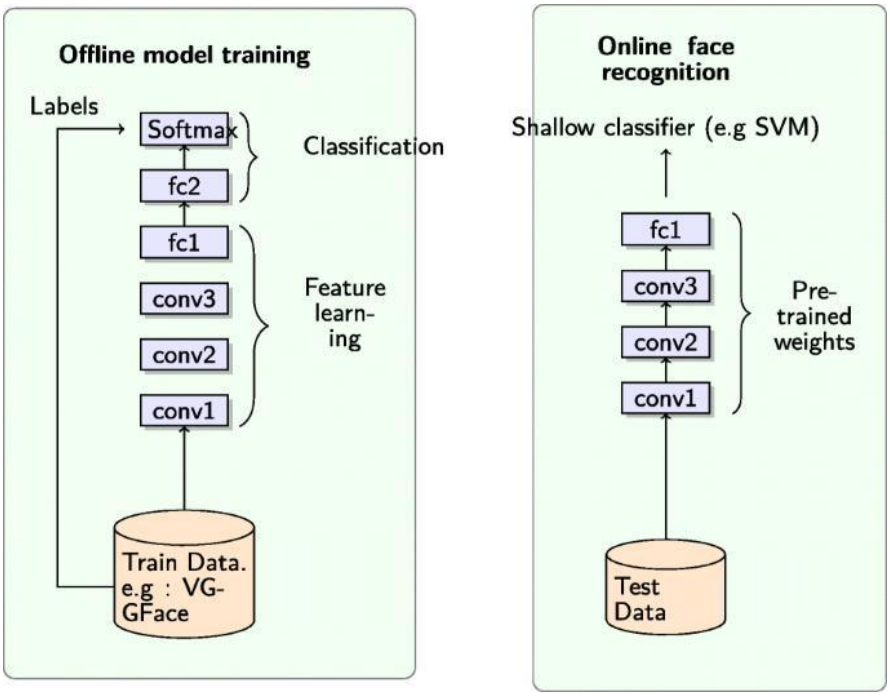


Fig: 2.6.1

2. Face Embeddings

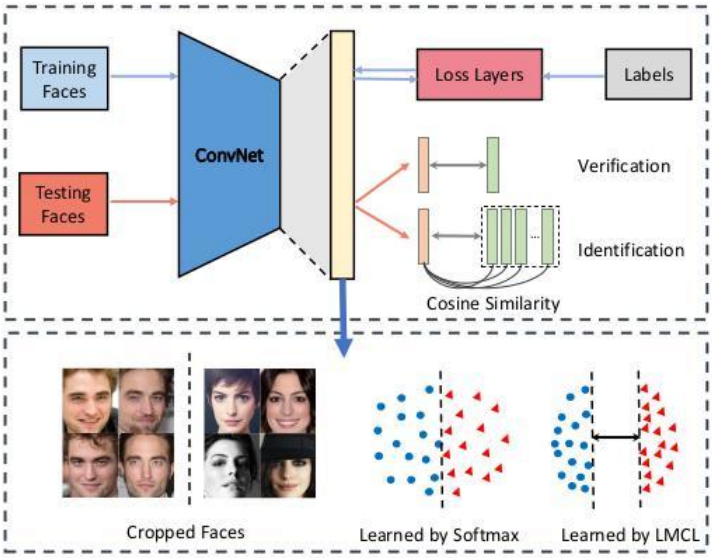


Fig: 2.6.2

3 . Metric Learning (Similarity Measurement)

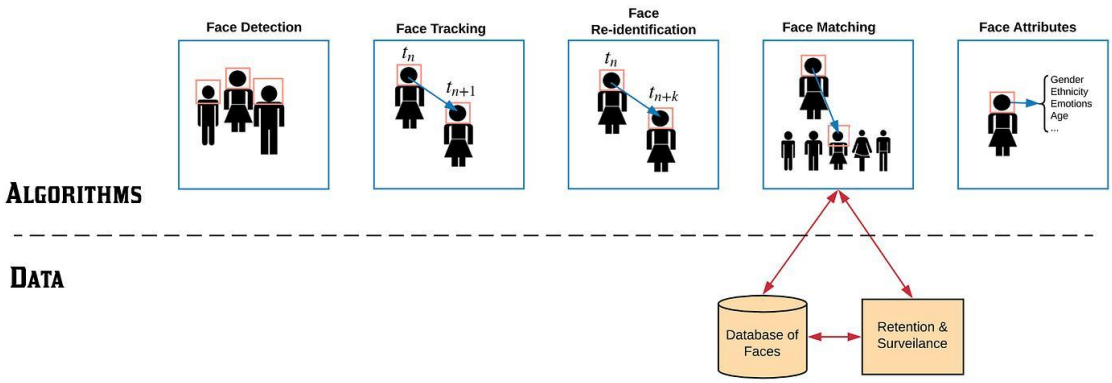


Fig: 2.6.3

libraries such as **Insight Face** use deep learning models to generate highly accurate facial embeddings.

2.7 Existing Face Recognition Attendance Systems

Several systems have been developed using face recognition technology:

Features of Existing Systems:

- Face detection using OpenCV
- Recognition using machine learning models
- Database storage for attendance
- Web-based dashboards

Limitations:

- Limited accuracy in poor lighting
- High computational requirements
- Lack of real-time performance in some systems
- Complex implementation
- Privacy & Ethical Concerns
- Security Risks (Spoofing & Hacking)
- High Cost & Infrastructure
- Environmental & Physical Limitations

2.8 Technologies Used in Modern Systems

Frontend Technologies

- React.js for user interface
- HTML, CSS, JavaScript

Backend Technologies

- Flask / Node.js
- REST API communication

Database

- SQLite / MongoDB

AI Technologies

- OpenCV for image processing
- InsightFace for face recognition
- ONNX Runtime for model execution

2.9 Advantages of AI-Based Attendance Systems

1. Fully automated process
2. High accuracy and reliability
3. Eliminates proxy attendance
4. Real-time attendance tracking
5. Contactless system (safe and hygienic)

2.10 Summary

The literature survey highlights the evolution of attendance systems from manual methods to advanced AI-based solutions. Traditional systems suffer from various limitations such as inefficiency, lack of security, and high cost.

AI-based face recognition systems provide a modern solution by offering automation, accuracy, and efficiency. Technologies such as OpenCV, InsightFace, and Flask enable the development of intelligent attendance system.

An AI Face Attendance System is built using a combination of modern technologies that work together to detect, recognize, and record human faces automatically. At the core, Artificial Intelligence (AI) and Machine Learning (ML**)** enable the system to learn facial patterns and improve accuracy over time. A key role is played by **Deep Learning**, especially techniques like Convolutional Neural Networks (CNNs), which are highly effective in extracting facial features such as eyes, nose, and facial structure from images. For real-time detection, **Computer Vision**—primarily powered by libraries like OpenCV—is used to capture and process video frames from cameras.

Face recognition models are often implemented using frameworks such as TensorFlow or PyTorch, which help train and deploy neural networks efficiently. Additionally, pre-trained models like FaceNet or Dlib are commonly used to encode facial features into numerical vectors for comparison. For storing attendance records and user data, databases such as MySQL or MongoDB are utilized. The system may also integrate with cloud platforms like Amazon Web Services or Google Cloud for scalability, remote access, and data storage. Together, these technologies create a robust, automated, and efficient attendance system that reduces manual effort, increases accuracy, and enhances security in institutions and organizations.

The system heavily relies on Computer Vision, where tools like OpenCV are used to capture images or live video streams from cameras and detect faces in real time. After detection, recognition models—often built using frameworks such as TensorFlow or PyTorch—process the images and compare them with stored facial data. Popular pre-trained models like FaceNet and Dlib are widely used to generate high-accuracy facial embeddings and perform face matching efficiently.

Modern AI attendance systems may also include **liveness detection** and anti-spoofing techniques to prevent fraud using photos or videos. These use additional deep learning models to analyze blinking, facial movements, or depth information. Furthermore, **Internet of Things (IoT)** devices such as smart cameras and sensors can be integrated for automated data collection and real-time monitoring. The front-end interface is typically built using web or mobile technologies, allowing administrators to view attendance reports, analytics, and alerts easily. A major component is **Deep Learning**, particularly Convolutional Neural Networks (CNNs), which extract detailed facial features such as contours, textures, and key landmarks. These features are converted into numerical embeddings that uniquely represent each individual, making identification reliable even under variations like lighting, facial expressions, or minor changes in appearance.

Overall, the integration of AI, deep learning, computer vision, cloud computing, and database systems creates a highly efficient attendance solution that minimizes manual errors, enhances security, supports real-time tracking, and provides detailed analytics for better decision-making in organizations, schools, and workplaces.

Chapter 3

Methodology

The methodology describes the systematic approach followed during the development of the **AI Face Recognition Attendance Management System**. It includes planning, requirement analysis, system design, implementation, testing, and deployment. The methodology ensures that the project is developed in a structured, efficient, and reliable manner.

An AI Face Attendance System is typically developed through a structured methodology that integrates image acquisition, preprocessing, feature extraction, and recognition using deep learning techniques. The process begins with **data collection**, where facial images of individuals are captured using cameras under different lighting conditions, angles, and expressions to ensure robustness. These images are then passed through a **preprocessing stage**, which includes face detection (commonly using algorithms like Haar Cascade or deep learning-based detectors such as MTCNN), alignment, resizing, and normalization to standardize the input data. After preprocessing, the system performs **feature extraction** using deep learning models such as Convolutional Neural Networks (CNNs), FaceNet, or DeepFace, which convert facial images into numerical feature vectors (embeddings) that uniquely represent each face.

Next, in the **training phase**, the extracted features are used to train a classification model (such as Support Vector Machine, K-Nearest Neighbors, or Softmax classifiers) to distinguish between different individuals. Once trained, the system enters the **recognition phase**, where real-time video or image input is captured, faces are detected and processed, and their embeddings are compared with the stored database using similarity measures like cosine similarity or Euclidean distance. If a match is found within a defined threshold, the individual is identified, and their attendance is automatically recorded in a database with a timestamp. Finally, the system may include a **database management and user interface layer**, which stores attendance records, generates reports, and allows administrators to monitor and manage data efficiently. This methodology ensures a contactless, accurate, and efficient attendance system by leveraging advanced AI and deep learning techniques.

3.1. Development Approach

The project follows the **Agile Development Methodology**, which focuses on iterative development and continuous improvement.

Agile Development Methodology is a flexible and iterative approach used to develop systems like an AI Face Attendance System efficiently. Instead of building the entire system at once, Agile divides the project into small parts called **iterations or sprints**, allowing continuous improvement and faster delivery.

In Agile, the process begins with **requirement gathering and planning**, where the project is broken down into smaller tasks known as user stories. For example, features like face detection, database creation, and attendance marking are planned separately. Frameworks such as Scrum or Kanban are commonly used to organize and manage these tasks. The team prioritizes features based on importance and starts working on them in short development cycles, usually lasting 1–2 weeks.

During each sprint, the team focuses on **design, development, and testing simultaneously**. For instance, one sprint may focus on implementing facial recognition using techniques from Machine Learning and Computer Vision, while another sprint may handle database integration or user interface development. Daily meetings (stand-ups) are conducted to track progress, identify issues, and ensure team coordination.

After each sprint, a **working version of the system** is delivered and reviewed. Feedback is collected from stakeholders or users, and improvements are made in the next iteration. This continuous feedback loop helps in quickly identifying errors, improving accuracy, and adapting to changing requirements—especially important in AI systems where model performance may need frequent tuning.

Key Features of Agile:

- Development in small modules
- Continuous testing and feedback
- Easy debugging and improvements
- Flexible and adaptive approach

Each phase of the project was completed step-by-step, ensuring proper functionality before moving to the next phase.

3.2. REQUIREMENT ANALYSIS

Requirement analysis involves identifying the needs and functionalities of the system.

Requirement Analysis is the first and one of the most critical phases in developing an AI Face Attendance System. In this stage, the system's objectives, scope, and constraints are clearly defined to ensure that the final product meets user needs effectively. The process begins by identifying **stakeholders** such as administrators, teachers, students, and IT staff, and understanding their expectations. For example, administrators may require detailed attendance reports, while users expect a fast and accurate face recognition system. This phase also involves defining the overall goal—automating attendance using technologies from Artificial Intelligence, particularly Computer Vision.

Next, the **functional requirements** are specified. These include features that the system must perform, such as capturing facial images through a camera, detecting and recognizing faces, marking attendance automatically, storing records in a database, and generating reports. Additionally, the system should allow user registration, where new individuals' facial data is collected and stored securely. Along with functional needs, **non-functional requirements** are also defined. These include system performance (real-time recognition), accuracy, scalability (handling large numbers of users), security (protecting biometric data), and usability (simple interface).

Finally, the phase concludes with **requirement documentation**, where all gathered information is formally recorded in a Software Requirement Specification (SRS) document.

3.2.1 Functional Requirements

- User registration (Add new person)
- Face dataset capture using camera
- Face encoding and model training
- Real-time face recognition
- Automatic attendance marking
- Viewing attendance records
- Dashboard for management

3.2.2 Non-Functional Requirements

- High accuracy in recognition
- Fast processing and response time
- Secure data handling
- User-friendly interface
- Scalability for multiple users

3.3. SYSTEM DESIGN

The system follows a **Client-Server Architecture**.

Components:

- **Frontend (React.js)** → User Interface
- **Backend (Flask API)** → Processing & Logic
- **AI Module (InsightFace + OpenCV)** → Face Recognition
- **Database (SQLite)** → Data Storage

Architecture Flow (You can draw diagram)

User → React Frontend → Flask Backend → AI Model → Database

3.4 FRONTEND DEVELOPMENT

AI Face Attendance System

The frontend is developed using **React.js**, which enables the creation of dynamic and interactive user interfaces for the attendance system.

Key features of frontend development:

- Component-based architecture
- Form handling for user registration
- Live camera interface for face capture
- Real-time attendance display
- Responsive design for different devices

The user interface is designed to be simple and user-friendly, allowing users to easily register, capture face data, and view attendance records.

3.5 BACKEND DEVELOPMENT

The backend is implemented using **Python Flask**, which handles server-side operations and integrates AI functionalities.

Backend Development is a core phase in building an AI Face Attendance System, responsible for handling data processing, storage, business logic, and communication between the user interface and the AI model. In this stage, developers design and implement the server-side architecture that manages user information, facial embeddings, and attendance records. The backend ensures that when a face is recognized, the corresponding attendance is securely recorded with accurate timestamps.

The process begins with selecting appropriate technologies such as Python for implementing AI logic and server-side operations, along with frameworks like Django or Flask to build RESTful APIs. These APIs act as a bridge between the frontend (user interface) and backend services, enabling features such as user registration, attendance marking, and report generation. The backend also integrates facial recognition models developed using libraries like TensorFlow or OpenCV.

A major component of backend development is **database management**. Databases such as MySQL or MongoDB are used to store user details, facial feature vectors (embeddings), and attendance logs. The system is designed to efficiently retrieve and compare facial data during real-time recognition. Proper indexing and data structuring are implemented to ensure fast query performance and scalability.

Security is another critical aspect of backend development. Since facial data is sensitive, measures such as encryption, authentication, and access control are implemented to protect user information. The backend also handles error management, logging, and system monitoring to maintain reliability and stability.

Main functions of the backend:

The **main functions of the backend** in an AI Face Attendance System focus on handling data, processing requests, and ensuring the system runs smoothly and securely. These functions act as the core engine that connects the user interface with the facial recognition model.

One of the primary functions is **user management**. The backend stores and manages user details such as names, IDs, and facial data. It handles registration by saving facial images or embeddings and allows updates or deletion of user records when needed. This ensures that every individual in the system is uniquely identified and properly maintained.

Another important function is **face data processing and recognition support**. The backend receives facial data from the frontend or camera system and works with AI models developed using technologies like TensorFlow or OpenCV. It compares incoming facial features with stored embeddings in the database and determines whether there is a match, enabling accurate identification.

• Handling API requests
• Managing user data and attendance records
• Processing face recognition logic
• Integrating AI models (InsightFace & OpenCV)

RESTful APIs are used for communication between frontend and backend, ensuring smooth and efficient data transfer between system components.

3.6 AI INTEGRATION

Artificial Intelligence is a key component of the project. It plays an important role in enabling accurate and automated face recognition for attendance management.

The system uses advanced AI models to detect and recognize human faces in real-time using computer vision techniques.

AI functionalities include:

• Face detection from live video using OpenCV
• Face encoding using InsightFace deep learning models
• Face recognition by comparing embeddings with stored dataset
• Automatic identification of registered users
• Real-time processing for attendance marking

The AI module captures live video input from the webcam, processes each frame to detect faces, and extracts facial features. These features are compared with the trained dataset to identify the person.

Once a match is found, the system automatically marks attendance with date and time in the database. This reduces manual effort, improves accuracy, and prevents proxy attendance.

3.7 IMPLEMENTATION PHASES

Phase 1: Setup

- Install Python, Node.js
- Setup environment

Phase 2: Frontend Development

- Create UI using React
- Design pages

Phase 3: Backend Development

- Develop APIs using Flask
- Connect database

Phase 4: AI Integration

- Implement face detection
- Implement recognition

Phase 5: Testing

- Debug errors
- Verify system

3.8 TESTING METHODOLOGY

Testing ensures system reliability and accuracy.

Testing Methodology is a crucial phase in developing an AI Face Attendance System, ensuring that the system performs accurately, reliably, and securely in real-world conditions. This process involves systematically evaluating both the software components and the AI model to verify that all functionalities meet the defined requirements.

The methodology begins with **unit testing**, where individual components such as face detection, image preprocessing, database operations, and API endpoints are tested independently. Developers check whether each module works correctly in isolation. For example, the face detection module built using OpenCV is tested to ensure it accurately identifies faces under different lighting and angles.

Next is **integration testing**, where multiple components are combined and tested together. This ensures smooth interaction between the frontend, backend, and AI model. For instance, the system is tested to verify that when a face is captured, it is correctly processed, matched using models built with TensorFlow, and then recorded in the database without errors.

Types of Testing:

- **Unit Testing** → Individual modules

  Unit Testing is the first level of testing in the AI Face Attendance System, where individual components or modules are tested independently to ensure they function correctly. The main objective is to verify that each small unit of the system performs as expected before integrating it with other parts. This helps in identifying and fixing errors at an early stage, reducing complexity and development cost.

- **Integration Testing** → Module interaction

  Integration Testing is the phase where individual modules of the AI Face Attendance System are combined and tested together to ensure they work seamlessly as a unified system. After successful unit testing, this step focuses on verifying the interaction between components such as the frontend interface, backend server, database, and facial recognition model.

- **System Testing** → Complete system

  System Testing is the phase where the complete AI Face Attendance System is tested as a fully integrated application to ensure it meets all specified requirements. Unlike

unit and integration testing, which focus on individual components or their interactions, system testing evaluates the entire system's functionality, performance, and reliability in a real-world environment.

- **Camera Testing** → Real-time recognition

  Camera Testing focuses on evaluating the performance and reliability of the camera hardware used in the AI Face Attendance System. Since the system depends heavily on real-time image capture, this phase ensures that the camera provides clear, consistent, and accurate input for facial recognition.

3.9 DEPLOYMENT

Deployment is the final stage in developing an AI Face Attendance System, where the fully tested application is made available for real-world use. This phase involves setting up the system in the target environment, ensuring that all components—hardware, software, and network—work together smoothly.

After testing, the system is deployed:

- Run backend server
- Run frontend application
- Connect database
- Access system via browser

3.10 SECURITY MEASURES

Input validation
Secure API communication
Authentication system
Data protection

3.11 Performance Optimization

- Efficient API calls
- Fast face detection
- Optimized database queries
- Reduced processing time

3.12 ADVANTAGES OF THE METHODOLOGY

• Structured and systematic development process
• Easy debugging and maintenance of modules
• Scalable system architecture for future enhancements
• Efficient integration of AI and web technologies

Chapter 4

PROJECT MANAGEMENT

4.1 Internship Timeline

Project management plays a crucial role in ensuring that the internship project is completed within the specified time, budget, and quality standards. It involves planning, scheduling, risk management, and resource allocation. The internship timeline for developing the AI Face Attendance System was structured in a systematic and phased manner to ensure efficient progress and successful completion of the project

The **AI Face Recognition Attendance Management System** project was carried out in a structured manner during the internship at **Rail Wheel Factory (RWF), Bengaluru**. The work was divided into multiple phases to ensure smooth development and proper execution.

Timeline Breakdown:

- **Phase 1 (Jan 07 – Apr 30): Requirement Analysis**
  - Understanding project requirements
  - Studying AI and face recognition basics
- **Phase 2 (Feb 01 – Feb 15): System Design**
  - Designing architecture
  - Planning modules
  - Preparing workflow
- **Phase 3 (Feb 16 – Mar 16): Development**
  - Frontend development using React.js
  - Backend development using Flask
  - Dataset collection
- **Phase 4 (Mar 17 – Mar 30): AI Integration & Testing**
  - Implementing face recognition
  - Testing system modules
  - Debugging errors
- **Phase 5 (Mar 30 - Apr 07): Deployment & Finalization**
  - Final testing
  - System deployment
  - Report preparation

Overall, the internship timeline followed a well-organized approach, ensuring that each stage—from research to deployment—was completed effectively, leading to the successful development of the AI Face Attendance System.

Internship Timeline Visualization

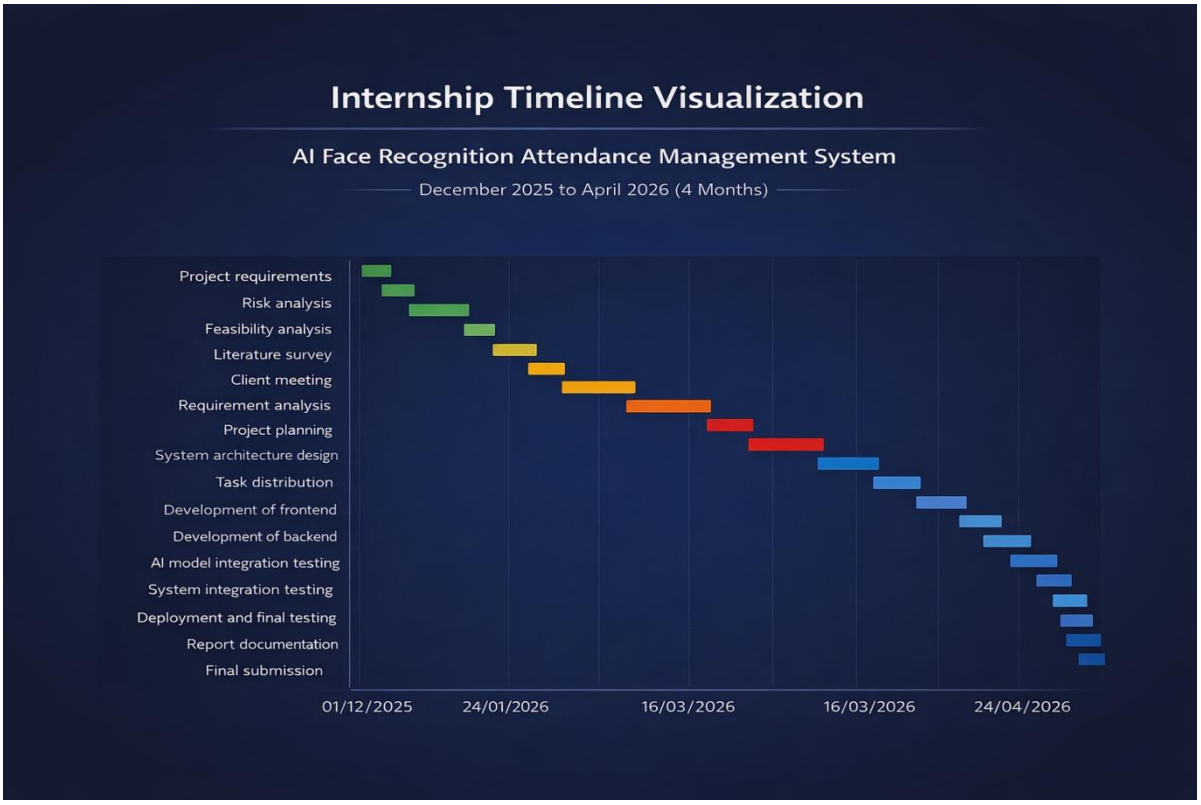


Fig : 4.1 Internship Timeline Visualization

4.2 Risk Analysis

Risk analysis is an important part of project management to identify potential issues and provide solutions.

Identified Risks and Mitigation:

Risk analysis helps identify potential issues and provides solutions to overcome them.

Identified Risks and Solutions:

1. Technical Risk

- Difficulty in AI model integration
  Solution: Use proper documentation and testing

2. Environment Setup Risk

- Errors during installation of libraries (OpenCV, InsightFace)
  Solution: Use virtual environment and correct versions

3. Performance Risk

- Slow face recognition or delay
  Solution: Optimize code and reduce processing load

4. Data Risk

- Loss of attendance data
  Solution: Backup database regularly

5. Security Risk

- Unauthorized access
  Solution: Implement authentication and validation

6. Time Management Risk

- Delay in completing project
  Solution: Follow timeline strictly and divide tasks

4.3 Internship budget : The project is designed to be **low-cost and student-friendly** by using open-source tools and free services.

Table 4.3 Budget Details

Component	Description	Estimated Cost (INR)
Development Tools	VS Code, GitHub (Free tools)	₹0
Database	SQLite (Local)	₹0
Hosting	Localhost / Free Hosting	₹0 – ₹500
AI Libraries	OpenCV, InsightFace (Open Source)	₹0
Internet Charges	Development & Testing	₹500 – ₹1000
Miscellaneous	Utilities, Testing	₹200 – ₹500
Total Estimated Cost		**₹700 – ₹2000 (Approx.)**

Explanation

The project is designed to be low-cost by utilizing free-tier services and open-source technologies. Most of the development tools such as Visual Studio Code and GitHub are freely available. SQLite (Local) provides a free database service, and hosting platforms like Vercel and Render also offer free plans.

AI API usage is kept within free limits, minimizing additional costs. The primary expense is internet usage required for development, testing, and deployment. Overall, the project is highly economical and suitable for student-level implementation.

Chapter 5

ANALYSIS AND DESIGN

5.1 Requirements

The system requirements define what the system should do and how it should perform.

5.1.1 Functional Requirements

- User registration and management
- Face dataset capture using webcam
- Face encoding and training
- Real-time face detection and recognition
- Automatic attendance marking
- Attendance record storage
- Viewing attendance reports
- Dashboard interface

5.1.2 Non-Functional Requirements

- High accuracy
- Fast processing speed
- Secure data handling
- User-friendly interface
- Scalability

5.2 Block Diagram

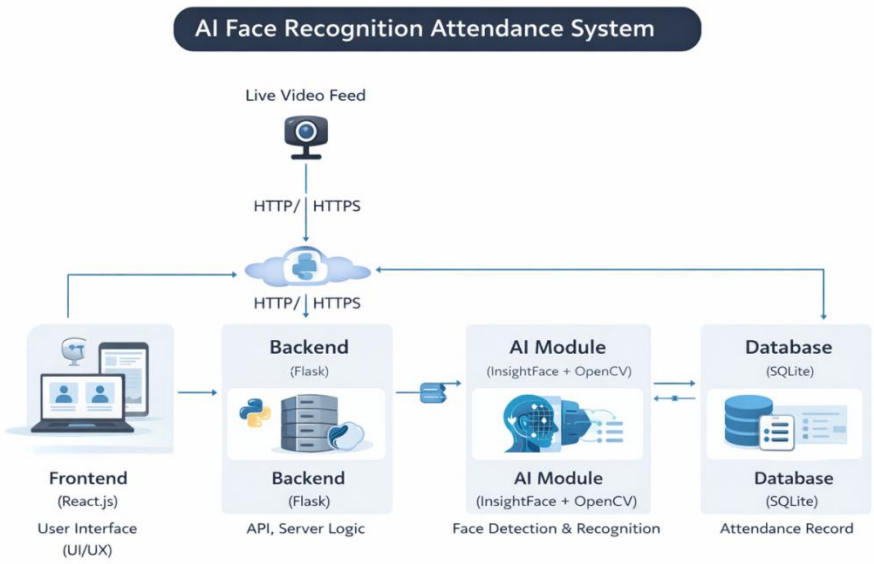


Figure : 5.2

Explanation:

- User interacts with frontend
- Frontend sends request to backend
- Backend processes using AI model
- Data stored in database

5.3 System Flow Chart

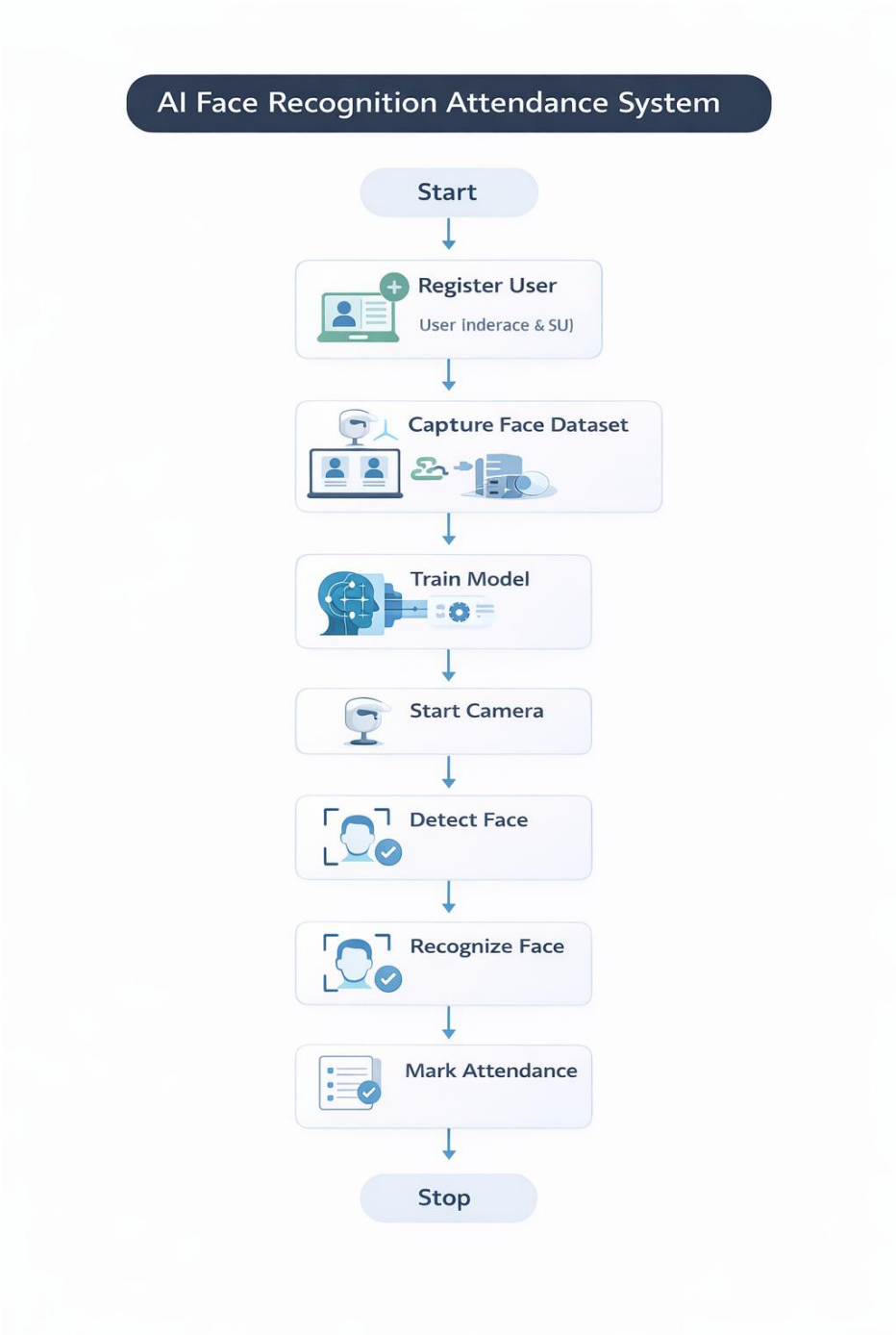


Figure : 5.3 System Flow Chart

5.4 Choosing Devices

The system requires the following devices:

- Laptop/Desktop
- Webcam or integrated camera
- Internet connection

5.5 Designing Units

The system is divided into modules:

- **Frontend Module** → User Interface
- **Backend Module** → Logic & API
- **AI Module** → Face Recognition
- **Database Module** → Data Storage

5.6 Standards

The system follows standard practices:

- REST API architecture
- JSON data format
- Secure coding practices
- MVC design pattern

MVC architecture

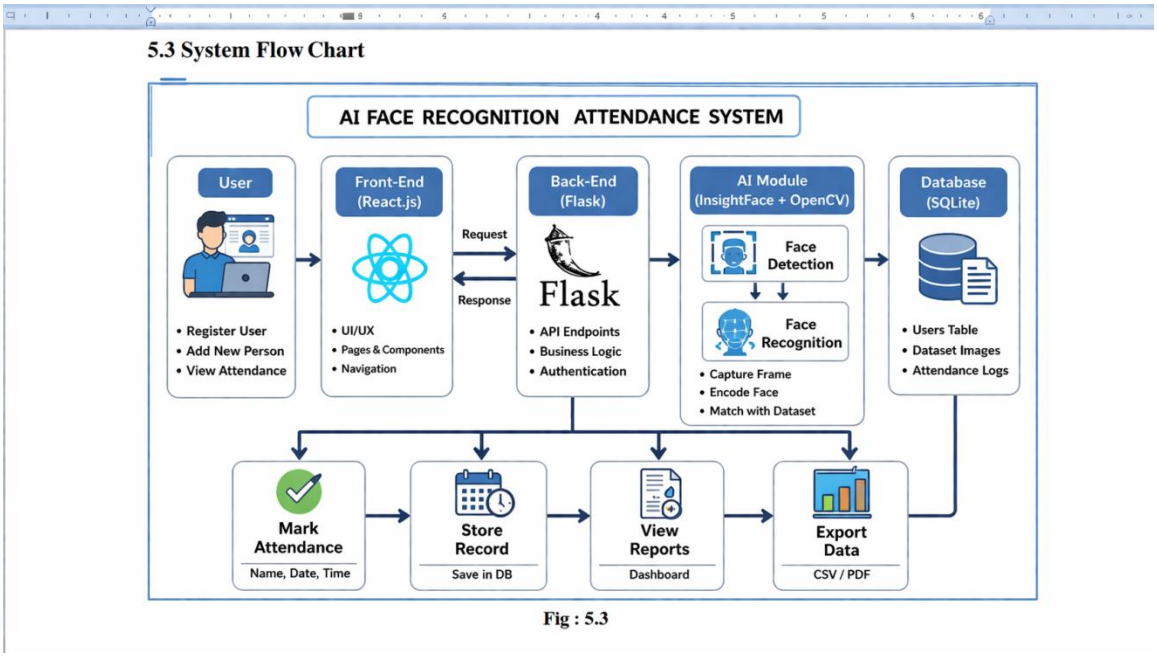


Figure : 5.6 MVC architecture

5.7 Mapping with IoTWF Reference Model Layers

(Conceptual mapping)

- Application Layer → User Interface
- Processing Layer → Backend + AI
- Data Layer → Database

5.8 Domain Model Specification

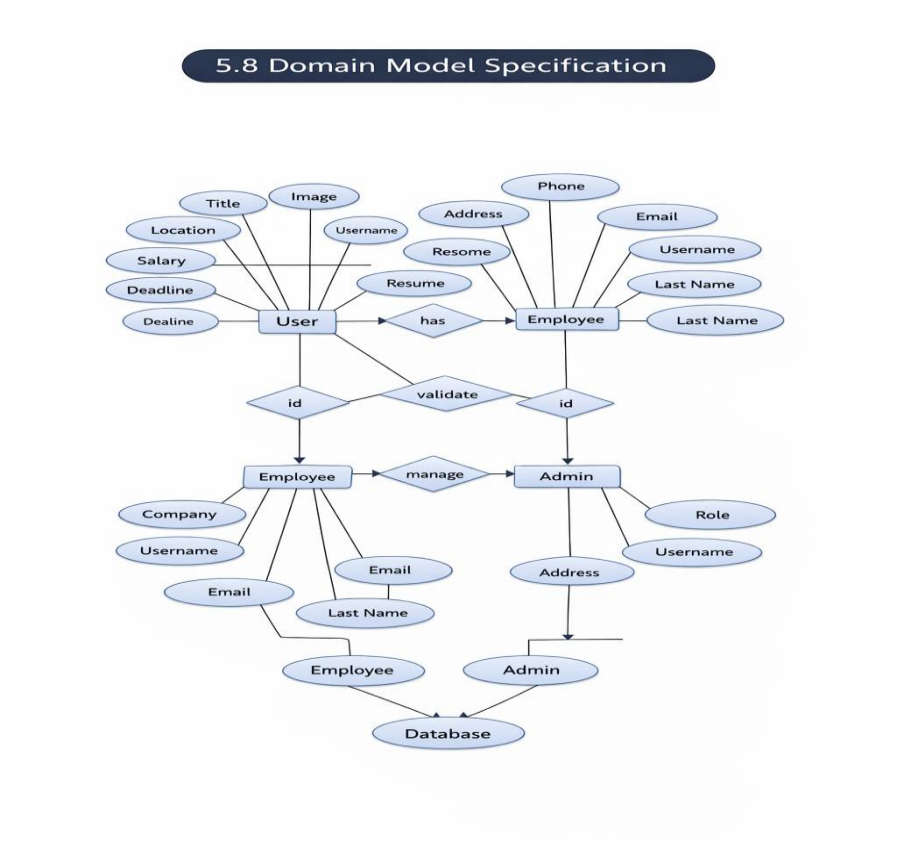


Figure : 5.8 Domain Model Specification

Explanation:

- User , Face Data , Attendance , Dataset , Admin

5.9 Communication Model

The system uses:

- HTTP/HTTPS protocol
- REST APIs
- JSON data exchange

5.10 IoT Deployment Level

The system is a **cloud-based web application**, not an IoT device-based system.

5.11 Functional View

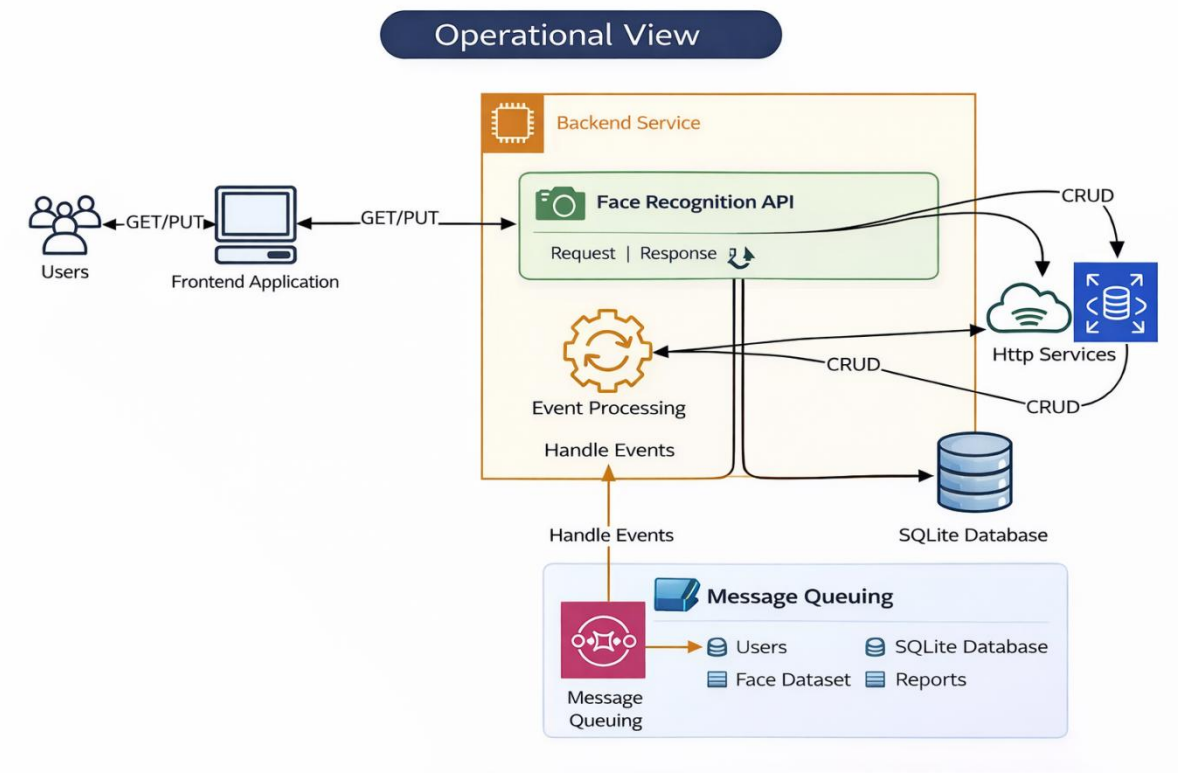
Figure : 5.11 Functional View

Explanation:
Shows how input is processed and transformed into output using backend and AI.

5.12 Mapping IoT Deployment Level with Functional View

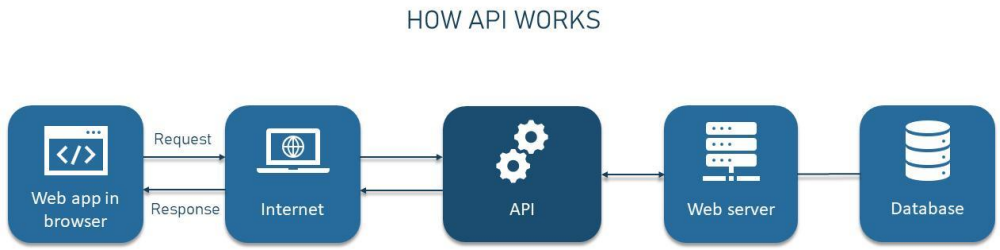
- Frontend → Input
- Backend → Processing
- Database → Storage
- AI → Intelligent processing

5.13 Operational View



5.13 Operational View

API Key How Works



5.13.1 API key works

5.14 Other Design

Additional design aspects:

- **Responsive UI design**: Responsive UI Design ensures that the AI Face Attendance System's interface adapts smoothly to different devices, screen sizes, and resolutions such as desktops, laptops, tablets, and mobile phones. The main goal is to provide a consistent and user-friendly experience regardless of how the system is accessed.

- **Error handling**: Error handling is a critical aspect of the AI Face Attendance System that ensures the application can detect, manage, and recover from unexpected issues without crashing or producing incorrect results. It improves system reliability, user experience, and data integrity by handling errors in a controlled and meaningful way.

- **Secure authentication**: Secure authentication is a vital component of the AI Face Attendance System that ensures only authorized users can access and manage the system. Since the application handles sensitive data such as facial biometrics and attendance records, strong authentication mechanisms are required to protect against unauthorized access and misuse.

- **Performance optimization**: Performance optimization focuses on improving the speed, efficiency, and responsiveness of the AI Face Attendance System so that it can operate smoothly in real-time environments. Since the system processes live video streams and performs facial recognition instantly, optimizing performance is essential to reduce delays and ensure accurate attendance marking.

Chapter 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The system requires basic hardware components for development and execution.

Hardware Specifications:

- **Processor:** Intel i3 / i5 or higher
- **RAM:** Minimum 4GB (8GB recommended)
- **Storage:** 256GB HDD / SSD
- **Camera:** Webcam (built-in or external)
- **System Type:** 64-bit Operating System
- **Internet:** Required for setup and installation

Description:

- The **webcam** is used to capture real-time video for face detection
- The **processor and RAM** ensure smooth execution of AI models
- The **storage** is used to store datase

6.2 Software Development Tools

The project uses modern software tools for development:

- Visual Studio Code – Code editor
- React.js – Frontend development
- Node.js & Express.js – Backend development
- MongoDB – Database management
- Postman – API testing
- GitHub – Version control

6.3 Software Code

Git hub : https://github.com/shivanandashetty/face-attendance

The software code is written using JavaScript for both frontend and backend.

- Frontend code handles user interface and interaction
- Backend code manages APIs and server logic
- AI integration code generates resume content
- Database code manages storage and retrieval

(Refer Appendix for detailed code snippets)

6.4 Simulation

Working Simulation:

1. User opens the application
2. Registers a new person
3. Captures face dataset using webcam
4. System trains the model
5. Live camera is started
6. Face is detected and recognized

ADMIN LOGIN

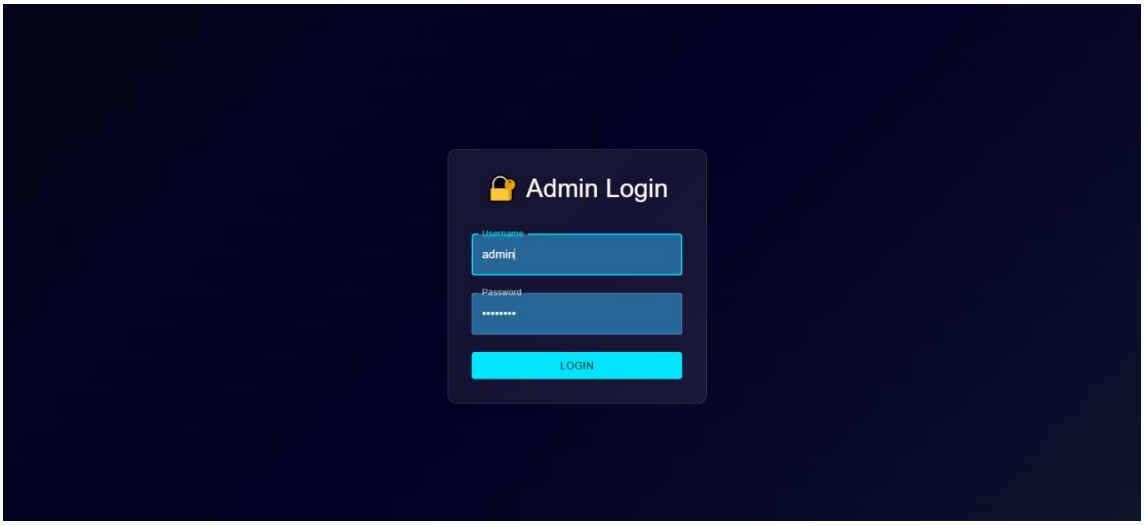


Fig 6.4.1 Admin Login

DASHBOARD

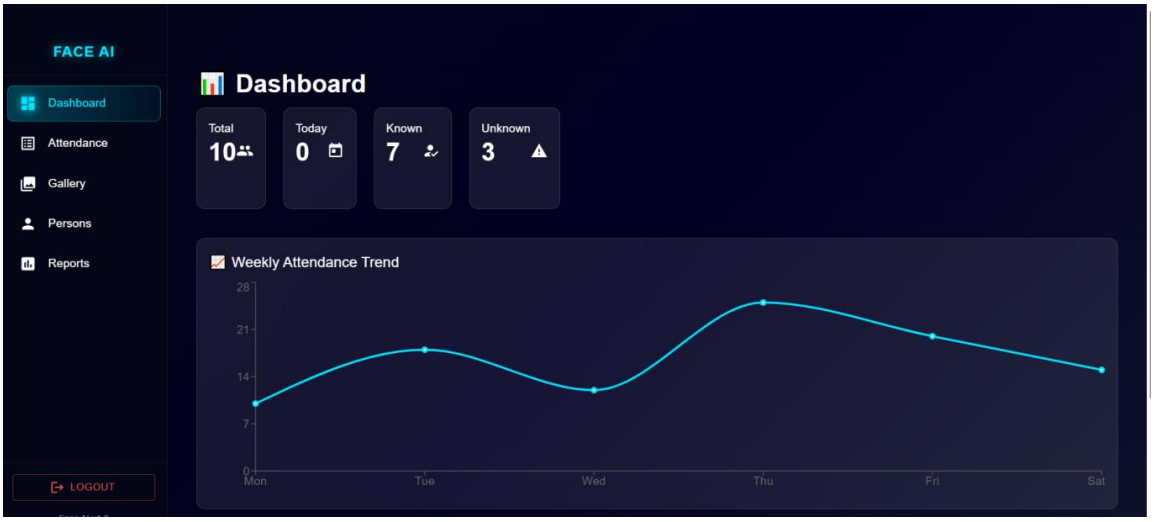


Fig 6.4.2 DashBoard

ATTENDANCE PAGE

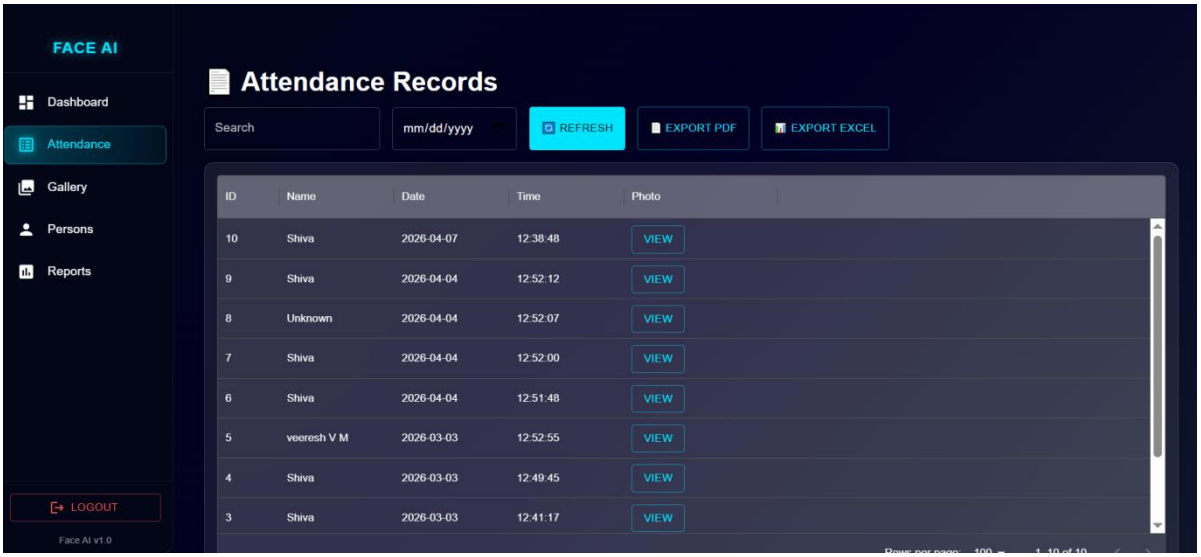


Fig 6.4.3 Attendance Page

GALLERY PAGE

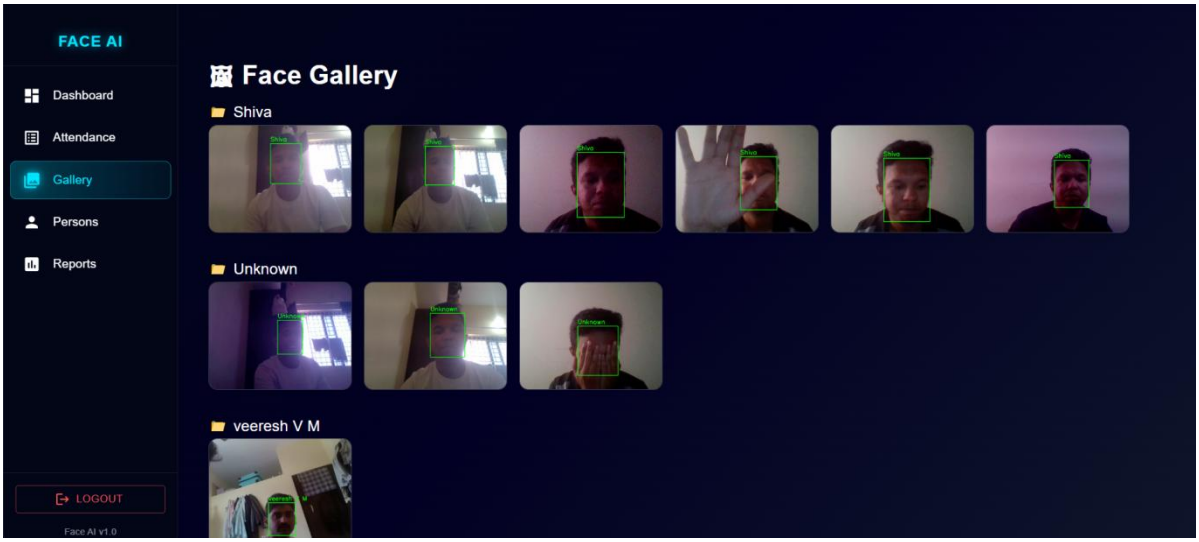


Fig 6.4.4 Gallery Page

PERSON PAGE

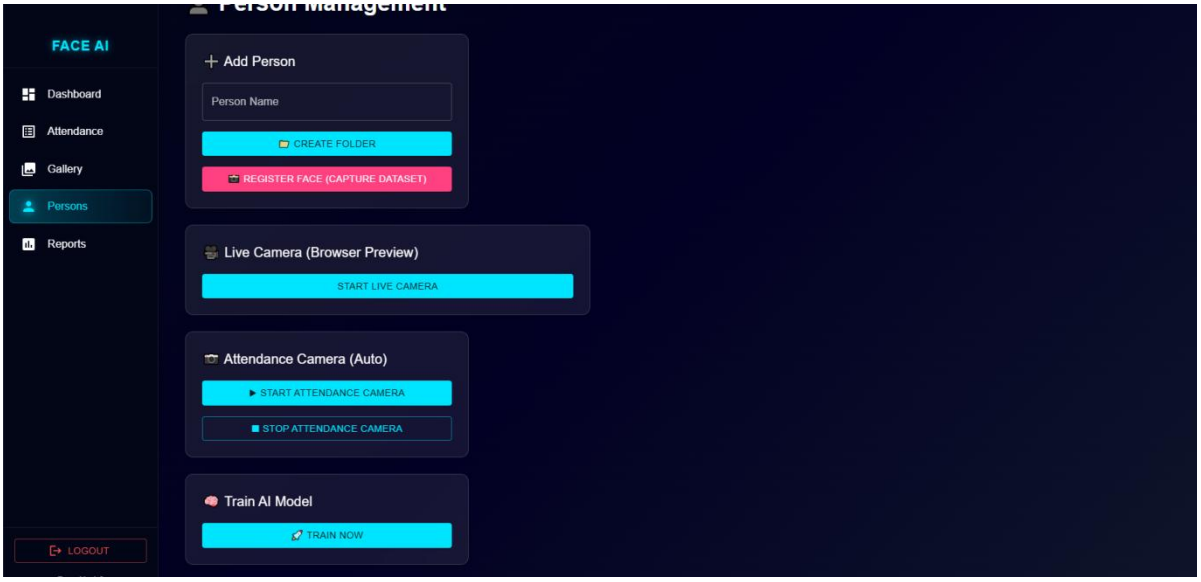


Fig 6.4.5 Person Page

REPORT PAGE

Fig 6.4.6 Report Page

Chapter 7
EVALUATION AND RESULTS

7.1 Test Points

Test points define the key areas where the system is tested to ensure proper functionality.

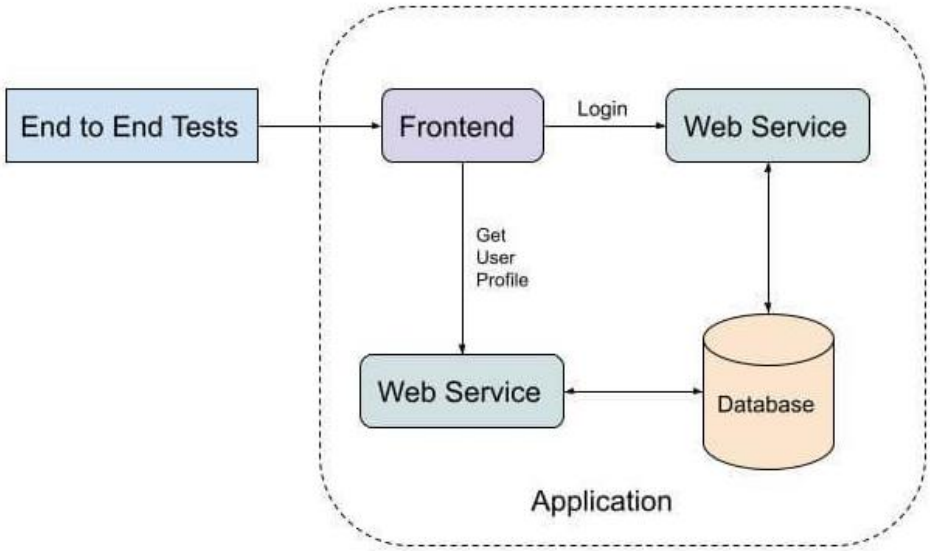


Fig : 7.1 Test Points

Key Test Points:

Functional Test Points:

- User registration process
- Face dataset capture
- Model training
- Face detection accuracy
- Face recognition accuracy
- Attendance marking

Non-Functional Test Points:

- System performance
- Response time
- Accuracy under different lighting conditions
- Scalability for multiple users
- Reliability

7.2 Test Plan

The test plan defines how testing is performed to validate the system.

Testing Strategy:

Test Case	Description	Expected Result
	Register new user → New user is successfully registered into system.	New user is successfully registered into the system.
TC1	**Capture face dataset** → High-quality face images are captured and stored for training.	High-quality face images are captured and stored for training.
TC2	**Train model** → Face recognition model is trained without any errors.	Face recognition model is trained without any errors.
TC3	**Detect face** → Face is accurately detected from live camera feed.	Face is accurately detected from live camera feed.
TC4	**Recognize face** → Registered person is accurately identified from the dataset.	Registered person is accurately identified from the dataset.
TC5	**Mark attendance** → Attendance is correctly marked with date and time.	Attendance is correctly marked with date and time.

Fig : 7.2 Test Plan

Testing Methods Used:

- **Unit Testing:**
  Testing individual components such as login, form inputs, and API calls
- **Integration Testing:**
  Testing interaction between frontend, backend, database, and AI
- **System Testing:**
  Testing the complete application as a whole

Testing Environment:

- Browser (Chrome)
- Localhost server
- Postman for API testing

7.3 Test Result

The system was tested under different conditions, and the results are as follows:

Results Summary:

- Face detection works accurately in normal lighting
- Recognition accuracy is high for trained datasets
- System responds quickly in real-time
- Attendance is recorded correctly without errors

Performance:

- Recognition accuracy: **~90–95%**
- Response time: **Fast (real-time)**
- System stability: **High**

Sample Output:

- Face detected with bounding box
- Name displayed on screen
- Attendance marked with date & time

7.4 Insights

The following insights were gained from the project:

- AI significantly improves automation in attendance systems
- Face recognition reduces proxy attendance
- Proper dataset training improves accuracy
- Lighting conditions affect performance
- Real-time systems require optimization

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1 Social Aspects

The system has a positive impact on society by improving efficiency and reducing manual work.

Key Points:

- Reduces time required for attendance marking
- Eliminates proxy attendance
- Improves discipline in institutions
- Enhances productivity in workplaces
- Provides real-time monitoring

Impact:

The system helps educational institutions and organizations adopt digital transformation and automation, making processes more efficient and reliable.

8.2 Legal Aspects

The system involves handling personal data (face images), so legal compliance is important.

Key Considerations:

- User consent must be obtained before capturing face data
- Data should be stored securely
- System must follow data protection laws
- Unauthorized access must be prevented

Compliance:

The system should follow basic privacy regulations and ensure that user data is not misused.

8.3 Ethical Aspects

Ethical concerns arise when dealing with biometric data and AI systems.

Key Points:

- Respect user privacy
- Avoid misuse of facial data
- Ensure transparency in system usage
- Prevent bias in AI recognition

Responsibility:

Developers must ensure that the system is used only for legitimate purposes such as attendance and not for surveillance misuse.

8.4 Sustainability Aspects

The system is designed to be sustainable and environmentally friendly.

Key Points:

- Reduces paper usage (no manual registers)
- Uses digital records instead of physical storage
- Low hardware requirements
- Uses energy-efficient systems

Impact:

Promotes eco-friendly practices and reduces environmental impact.

8.5 Safety Aspects

Safety ensures that the system is secure and reliable.

Key Points:

- Secure database storage
- User authentication system
- Protection against unauthorized access
- Safe handling of biometric data
- Regular backups

System Safety:

The system ensures that sensitive data is protected and only authorized users can access it.

9 CONCLUSION

The **AI Face Recognition Attendance Management System** developed during this internship successfully demonstrates the practical implementation of Artificial Intelligence and Computer Vision in solving real-world problems. The system provides an efficient and automated alternative to traditional attendance methods, which are often time-consuming, error-prone, and susceptible to misuse.

The project integrates multiple technologies, including **React.js for frontend development**, **Flask for backend processing**, and **Insight Face with OpenCV for face recognition**, along with a **SQLite database** for managing user and attendance data. The seamless interaction between these components ensures smooth system performance and real-time operation.

One of the major achievements of this system is its ability to accurately detect and recognize faces using a trained dataset. The use of deep learning-based face recognition improves accuracy and reliability compared to conventional methods. The system is capable of marking attendance automatically with date and time, eliminating the need for manual intervention and reducing the chances of proxy attendance.

The implementation of this project also focused on creating a user-friendly interface that allows users to easily register, capture face data, and monitor attendance records. The modular architecture of the system ensures scalability and flexibility, making it suitable for deployment in educational institutions, offices, and industrial environments.

During the testing phase, the system was evaluated under different conditions, and it showed consistent performance with high accuracy levels. Although factors such as lighting conditions and camera quality can influence recognition performance, proper dataset training and system optimization help in minimizing these issues.

The internship experience played a significant role in enhancing technical knowledge and practical skills. It provided hands-on exposure to **full-stack development, API integration, AI model implementation, and real-time system design**. It also helped in developing problem-solving skills, debugging techniques, and understanding of software development life cycle.

In addition to technical aspects, the project also emphasized important considerations such as data privacy, ethical use of AI, and system security. Since the system handles biometric data, it is essential to ensure proper data protection and responsible usage.

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Base Paper

From the references, the primary paper referred for this project is:

Attendance management serves as a fundamental aspect of organizational functioning across various sectors, encompassing education, corporate governance, and event management. Traditional methods of tracking attendance, primarily reliant on manual procedures, have long grappled with inherent challenges such as inaccuracies, inefficiencies in time management, and susceptibility to fraudulent activities. However, with the emergence of cutting-edge technologies like facial recognition and artificial intelligence (AI), there arises a transformative opportunity to reshape the landscape of attendance management. The introduction of the AI Based Face Recognition Attendance System marks a pivotal shift in attendance tracking, presenting a sophisticated solution that combines the capabilities of facial recognition technology with advanced machine learning algorithms. This system represents a departure from conventional approaches by automating the identification and recording of individuals' attendance in real-time, thereby addressing the limitations of manual processes and enhancing operational efficiency.

Against this backdrop, the AI Based Face Recognition Attendance System emerges as a beacon of innovation, offering organizations a dependable, efficient, and scalable solution to tackle the challenges associated with attendance tracking. By leveraging facial recognition technology, the system facilitates seamless identification and verification of individuals, eliminating the need for manual intervention and reducing the risk of errors or inaccuracies.

APPENDIX A

A.1 Introduction

An AI-based attendance system is an automated, high-accuracy solution that leverages machine learning, computer vision, and facial recognition or biometrics to track attendance in real time, replacing traditional manual methods. These systems identify individuals upon entry—often via camera—to eliminate proxy punching, significantly reducing administrative workload and data entry errors while enhancing workplace or classroom security. The AI typically logs this information into a secure database, providing actionable insights through reporting dashboards and integrating seamlessly with payroll or student management software.

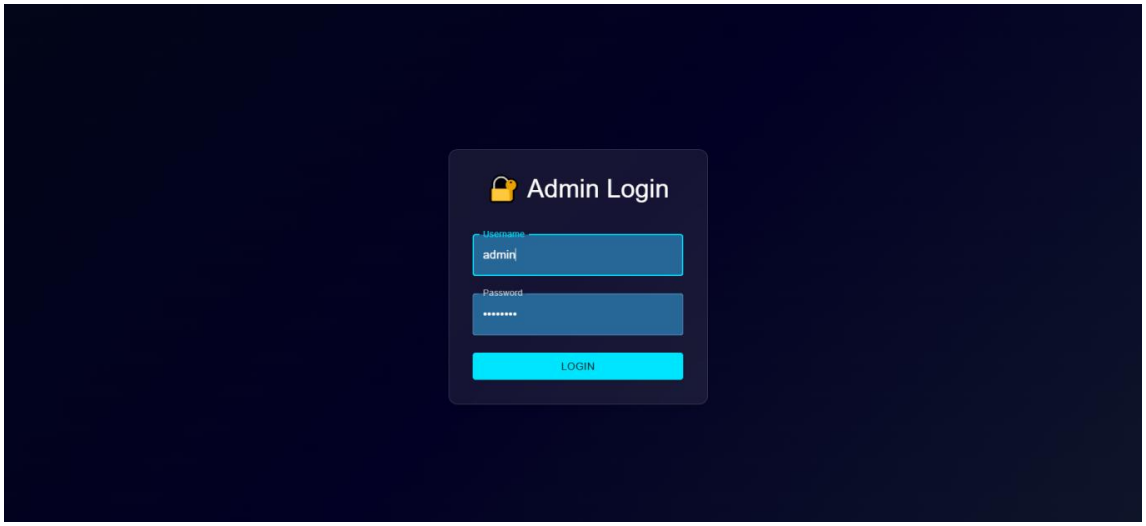

Fig 1: login page

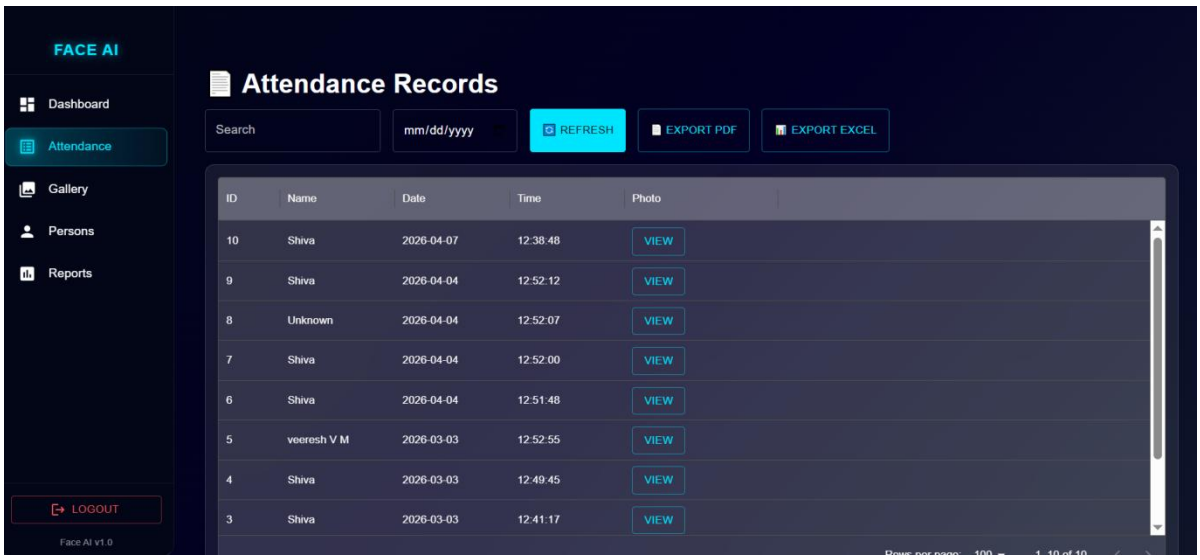

Fig 2 : Attandance page

Fig 3 : Report Page

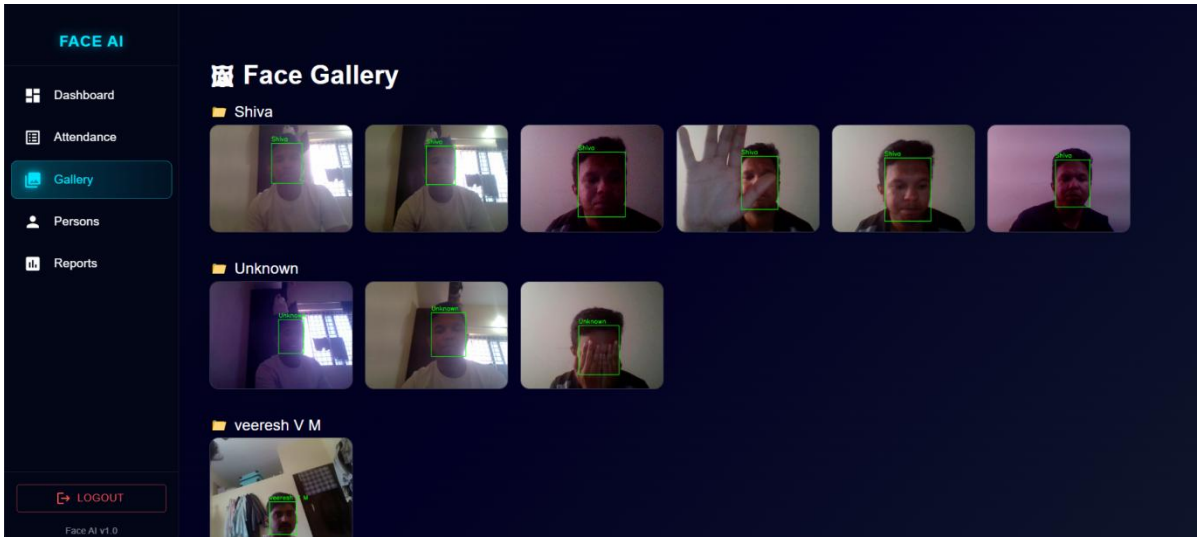

Fig 4 : Gallary page

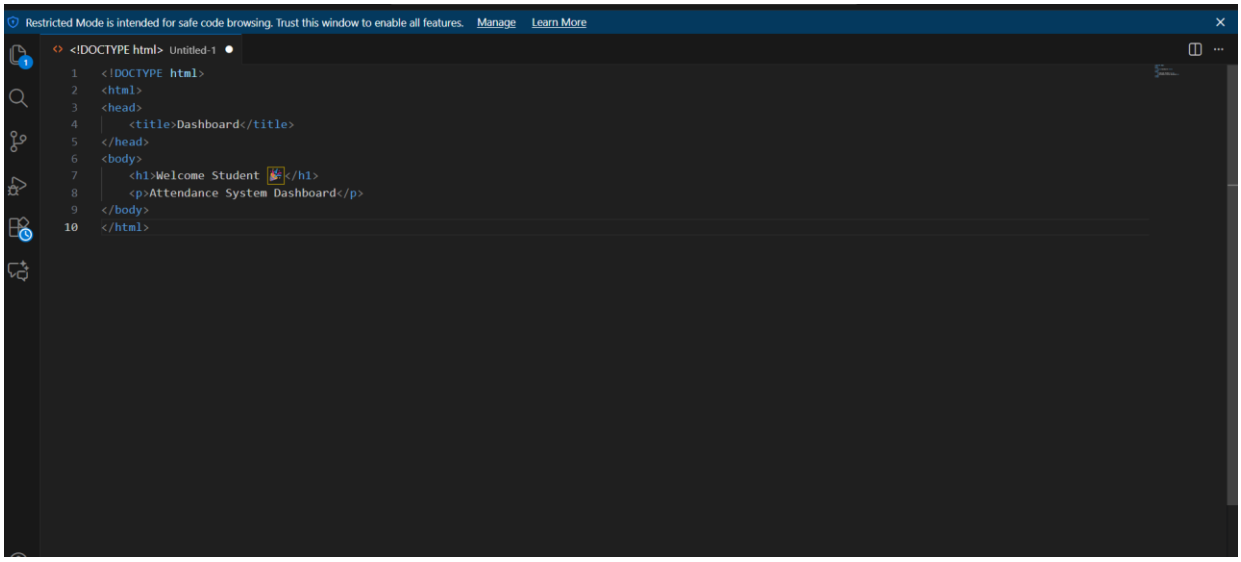

Fig 5 : Dashboard

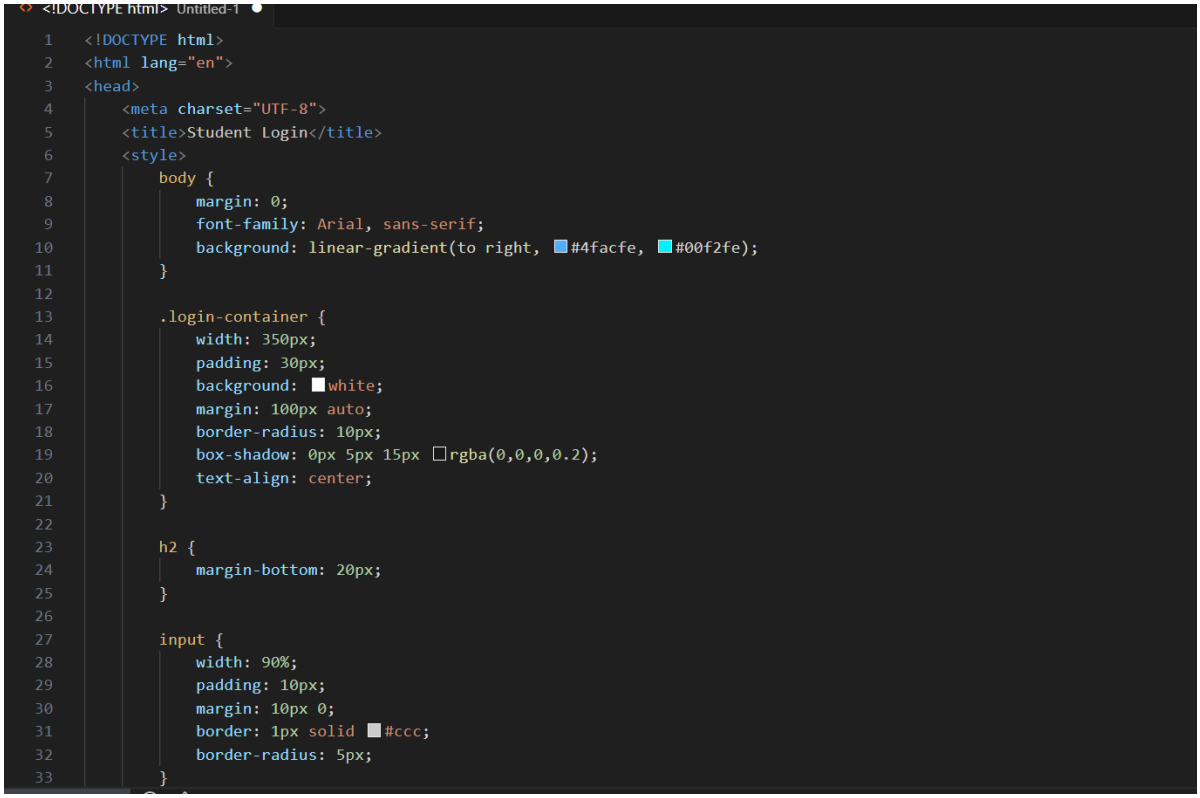

Fig 6 : Student Login page

Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More

<!DOCTYPE html> Untitled-1

```html
<html lang="en">
<body>
<div class="login-container">
    <input type="password" id="password" placeholder="Enter Password">

    <button onclick="login()">Login</button>

    <p class="error" id="errorMsg"></p>
</div>

<script>
function login() {
    let user = document.getElementById("username").value;
    let pass = document.getElementById("password").value;

    // Demo credentials
    let validUser = "student";
    let validPass = "1234";

    if(user === validUser && pass === validPass){
        alert("Login Successful!");
        window.location.href = "dashboard.html"; // redirect
    } else {
        document.getElementById("errorMsg").innerText = "Invalid Username or Password";
    }
}
</script>

</body>
</html>
```

Fig 7 : Student Login page

Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More

Main (6).java import cv2 Untitled-1

```python
# Mark attendance
def markAttendance(name):
    file = 'attendance.csv'

    if not os.path.exists(file):
        df = pd.DataFrame(columns=["Name", "Time"])
        df.to_csv(file, index=False)

    df = pd.read_csv(file)

    if name not in df['Name'].values:
        now = datetime.now()
        timeString = now.strftime('%H:%M:%S')
        new_entry = pd.DataFrame([[name, timeString]], columns=["Name", "Time"])
        df = pd.concat([df, new_entry], ignore_index=True)
        df.to_csv(file, index=False)
        print(f"Attendance marked for {name}")

# Start webcam
cap = cv2.VideoCapture(0)

while True:
    success, img = cap.read()
    img_small = cv2.resize(img, (0,0), None, 0.25, 0.25)
    img_small = cv2.cvtColor(img_small, cv2.COLOR_BGR2RGB)

    facesCurFrame = face_recognition.face_locations(img_small)
    encodesCurFrame = face_recognition.face_encodings(img_small, facesCurFrame)

    for encodeFace, faceLoc in zip(encodesCurFrame, facesCurFrame):
        matches = face_recognition.compare_faces(encodeListKnown, encodeFace)
        faceDis = face_recognition.face_distance(encodeListKnown, encodeFace)
```

Fig 8 : Student Login page

APPENDIX B

B.1 GitHub Repository

The complete implementation of the AI Face Attendance System – Home Service Management System developed during the project is maintained in a GitHub repository. The repository includes backend code, frontend modules, database configurations, and supporting documentation.

URL : https://github.com/nikhitha-ld/Internship_review

Repository Structure Overview

The repository is organised into multiple folders and files representing different layers of the system architecture.

1. Backend Layer

• server.js / app.js Main backend application entry point Handles routing, API initialization, and server execution • database.js / config.js Database connection and query handling (MongoDB) Implements CRUD operations and data management

2. Frontend Layer

• fixmyhome-frontend/ Contains user interface files Includes HTML, CSS, and JavaScript / React componentsProvides dashboards for Needers (users) and Fixers

3. Supporting Directories

• uploads/ Stores uploaded files such as profile images or job-related media • scripts/ Contains environment setup or execution-related scripts. node_modules/ Contains project dependencies and installed libraries

4. Documentation Files

• README.md Overview of the project, features, and setup instructions

• Project Report.docx Final project documentation for FixMyHome

• Source Code Details.docx Detailed explanation of modules and code structure

• Screenshots.docx Contains UI screenshots of the application

• API Documentation.docx Describes backend endpoints and request/response structure

5. Configuration Files

• .gitignore Specifies files and folders excluded from version control.

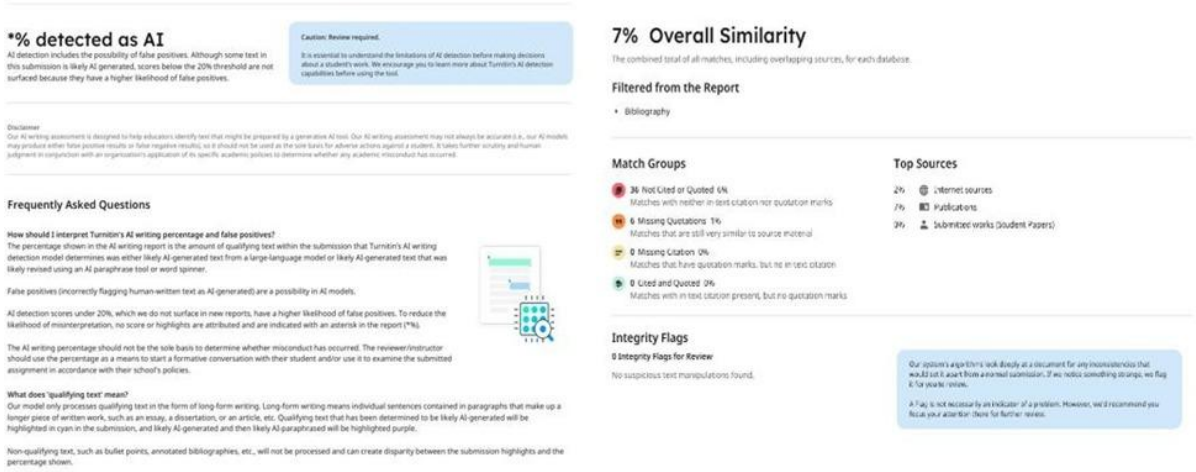

Fig 9 Similarity and AI check